

Hind Mukhtar

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Education

University of Ottawa, PhD in Electrical & Computer Engineering, AI-Enabled Wireless Communications	Sept 2021 – Sept 2026
University of Ottawa, Master in Electrical & Computer Engineering	Sept 2019 – May 2021
Queen's University, Bachelor in Electrical Engineering	Sept 2014 – May 2019
Licensed Professional Engineer (P.Eng), Ontario	Dec 2024 – Present

Technical Skills

Programming: Python, SQL, C/C++ (familiar)
ML Frameworks: PyTorch, Tensorflow
LLMs & GenAI: Google Gemini, Abacus AI, RAG, Agent Orchestration, Prompt Engineering
Data Visualization: Power BI, Tableau, Plotly, Dash, SQL Server Reporting Services (SSRS)
Cloud Services: AWS (EC2 GPU, Athena, Redshift, S3), Azure (Functions, DevOps)

Experience

Machine Learning Engineer , Gogo Business Aviation	Jan 2026 – Present
- Developed Text-to-SQL AI agents to enable natural language querying of enterprise datasets, empowering business users to self-serve analytics, reducing reporting development time. - Developed a RAG-powered sales assistant, integrating Salesforce, AWS Athena, and internal documentation to provide customer insights, retrieve pricing information, generate sales quotes and answer customer-related questions. - Developed cloud-hosted connectivity performance monitoring platforms for in-flight Wi-Fi connectivity, analyzing router telemetry, client device activity and network performance to improve user connectivity experience through real time diagnostics and anomaly detection. - Applied large language models and multi-agent orchestration to automate fault triage and log analysis across high-volume operational network data, reducing manual investigation time during connectivity incident response. - Designed, bench-marked and evaluated ML models on 10M+ records using AWS GPU clusters, implementing reproducible training pipelines and improving predictive performance across large-scale datasets.	
Senior Data Scientist , Gogo Business Aviation	Jan 2023 – Jan 2026
- Led real-time monitoring architecture for LEO, LTE and 5G product launches, engineering spatial-temporal pipelines across satellite telemetry and geolocation data to power proactive fault detection. - Built multi-variable forecasting models incorporating historical demand, projected growth, and allocation constraints to inform global satellite bandwidth allocation and provider contract strategy. - Built SQL queries, reporting pipelines, and dashboards to support engineering and product decision-making using PowerBI, Tableau and Python.	
Hardware Engineer , Satcom Direct	June 2019 – Jan 2023
- Designed and optimized multi-layer RF circuit boards for satellite-based in-flight connectivity systems, including simulation and RF parameter tuning using ADS and Altium Designer. - Developed automated RF, Wi-Fi and wireless test workflows using Python and LabVIEW, evaluating signal integrity and system parameters across multi-functional engineering teams.	

Publications

Handover Optimization in LEO Satellite Networks using Online Decision Transformer, IEEE Meditcom	July 2026
QoS Prediction for Satellite-Based Avionic Communication Using Transformers, IEEE TMLCN	Jan 2026